Fiscal Year:	FY 2006	**Task Last Updated:**	FY 12/02/2005
PI Name:	Loll, Patrick J. Ph.D.		
Project Title:	Detergent Interactions Affecting Membrane Protein Crystallization: Analysis and Use in Screen Design		
Division Name:	Physical Sciences		
Program/Discipline:	BIOTECHNOLOGY		
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Structural biology		
Joint Agency Name:		**TechPort:**	No
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Zip Code:	19102-1192	**Congressional District:**	2
Comments:			
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Flight Program:			
Flight Assignment:			
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COI Name (Institution):	Wiencek, J. (University of Iowa)		
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Task Description:	The quantitative analysis of detergent-detergent interactions will be carried out using measurements of the second osmotic virial coefficient (B22) obtained from static light scattering using the reaction center protein from Rb. sphaeroides. In addition, dynamic light scattering, infrared spectroscopic imaging, specific volume measurements, and solubility studies will be used to extend our understanding of the crystallization behavior of OmpF porin. To test our crystallization screens, we will express, purify, and study the crystallization behavior of four to six new (as yet uncrystallized) membrane proteins. These proteins will be analyzed in at least 12 different detergents, under conditions previously identified as promising for crystal formation. The principal objectives are to extend our quantitative analysis of detergent-detergent interactions and their contribution to crystal nucleation and growth, and to test experimental crystallization screens that we have recently developed using a battery of hitherto uncrystallized membrane proteins. Protein structural information is critical for structure-based drug design, protein engineering, and basic studies of how living cells function. The most powerful technique to obtain this knowledge is x-ray crystallography; this method,		

	however, has been nearly useless for membrane proteins, since this class of molecules has proven extremely difficult to crystallize. We are developing rational methods to overcome these difficulties, based upon the manipulation of the interaction forces underlying the crystallization behavior of complexes of membrane proteins and surfactant. We expect that this work will place membrane protein crystallization on a more rational footing, and believe that once the crystallization bottleneck is removed, we will be able to determine the atomic structures of these extremely important molecules in a straight-forward manner.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The ability to rationalize crystallize membrane proteins will prove to be invaluable in biology and drug design.
Task Progress:	The B22 behavior of protein-free detergent micelles is a good predictor of the behavior of the corresponding protein-detergent complexes (PDCs) under similar conditions, suggesting that detergent B22 measurements can be used as a screening tool when crystallizing PDCs. Our results with OmpF porin and the reaction center protein from R. sphaeroides show that the application of SLS to the measurement of virial coefficient data in complex systems has fundamental problems. Specifically, the solutions used to crystallize membrane proteins typically contain at a minimum three macro-sized particulate species, namely protein-detergent complexes, protein-free detergent micelles, and macromolecular precipitating agents such as polyethylene glycol (PEG). Some progress was made on assessing the effects of PEG in simplified binary systems. If the micelle size varies, B22 measurements will not accurately reflect micelle-micelle forces. We therefore examined micelle size in a model detergent system, using small angle X-ray scattering and static and dynamic light scattering, assessing the effects of temperature, detergent concentration, and precipitant on B22 and micelle size. In the absence of PEG, decreases in B22 principally reflect increases in micelle-micelle attractive forces, and do not reflect significant changes in micelle size. In the presence of PEG, the apparent hydrodynamic radius of detergent micelles shows a similar dependence upon micelle concentration as in the absence of PEG, suggesting PEG does not effect significant changes in micelle size, but rather acts by enhancing interaction forces between micelles. Taken together, these results show that reasonable paths exist by which one might hope to extract B22 data from various light scattering techniques; however, it is unclear if such methods will ever yield rapid, high-throughput results. In order to facilitate crystallization screening in a moderate throughput manner, we first tested whether the microbatch-under-oil method, which is compatible with our existing Oryx crystallization robot, was compatible with the detergents used in our crystallization screen. Specifically, we tested whether these detergents would partition significantly into the oils used in these experiments; we reasoned that if significant partitioning did indeed occur, the resulting reduction in detergent concentration in the protein solution, while potentially useful in guiding the crytallization process, would be difficult to control and reproduce and would therefore likely be detrimental to crystallization efforts. Our results showed that for two commonly used detergents, octyl glucoside and Fos-choline-12, the losses of detergent from the aqueous phase were insignificant. We also tested several membrane proteins for their behavior in our crystallization screens. These proteins included a nucleotide transporter, a metal transporter, and several outer membrane proteins. Unfortunately, we were unable to follow up on preliminary results or finish the proposed levels of testing, as levels of funding for the final year were drastically reduced, owing to an abandonment of this area of science by NASA.
Bibliography Type:	Description: (Last Updated: 01/05/2005)